Eyelid Abnormalities:

Malposition and Shape

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Lecture Acknowledgement and Thanks:

Professor William C. Hefner

Pacific University



Today's Conditions

- Ectropian
- Entropian
- Trichiasis
- Eyelid Coloboma
- Epicanthus
- Dermatochalasis

Describe these Eye Lids



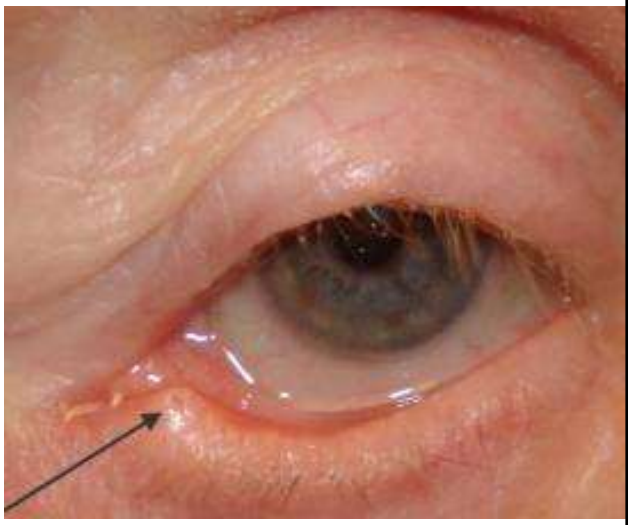
Ectropion

Abnormal eversion of lower lid margin away from globe



http://www.lensinkmd.com

Ectropion



- Ectropion can be described as:
  - Punctal
    - Can see puncta everted
  - Medial
    - Can see everted medial aspect of lid
  - Lateral
    - Can see everted lateral aspect of lid
  - Tarsal (complete)
    - Can see entire lower lid everted

Ectropion

Lower lid margin laxity gauged by:

- Lid distraction test
  - How far lid margin can be pulled away from globe
  - > 8mm = lid margin laxity
- Snap test
  - How quickly lid margin snaps back against globe after pulled away from it
    - > 1-2 seconds indicates lid margin laxity



Ectropion



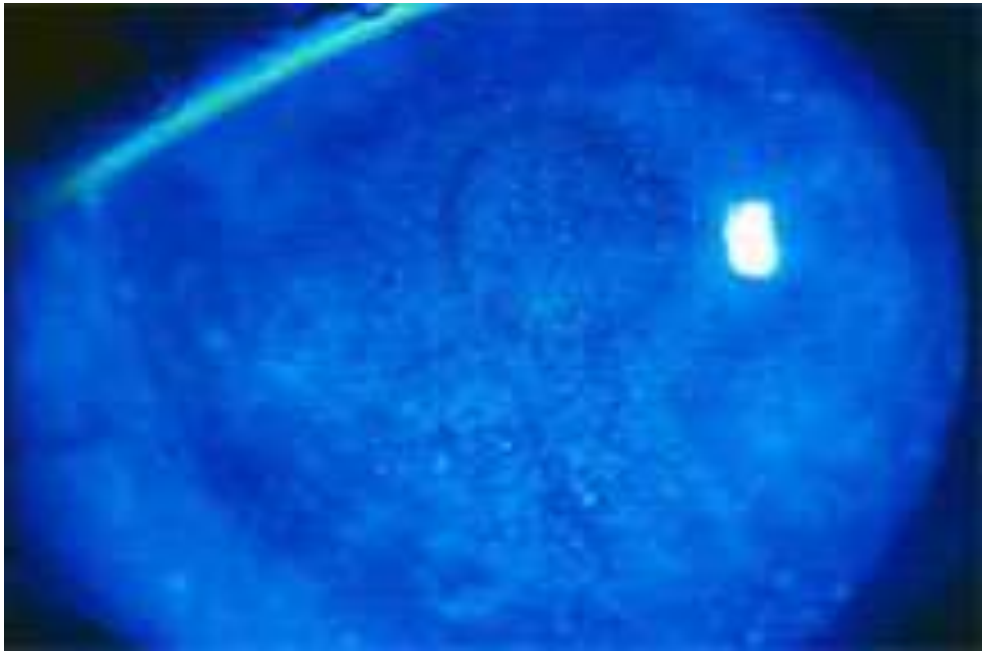
Subjective:

- Cosmesis
- Tearing
- Eye or eyelid irritation
- Constantly wiping eyes (exacerbates problem)
- History of burns, surgery or trauma around lids
- Concurrent Bell's Palsy
- Asymptomatic

Ectropion

Objective:

- Outward turning of eyelid margin
  - Associated corneal involvement:
    - Punctate Epithelial Erosions (PEE)
      - Corneal epithelium loss

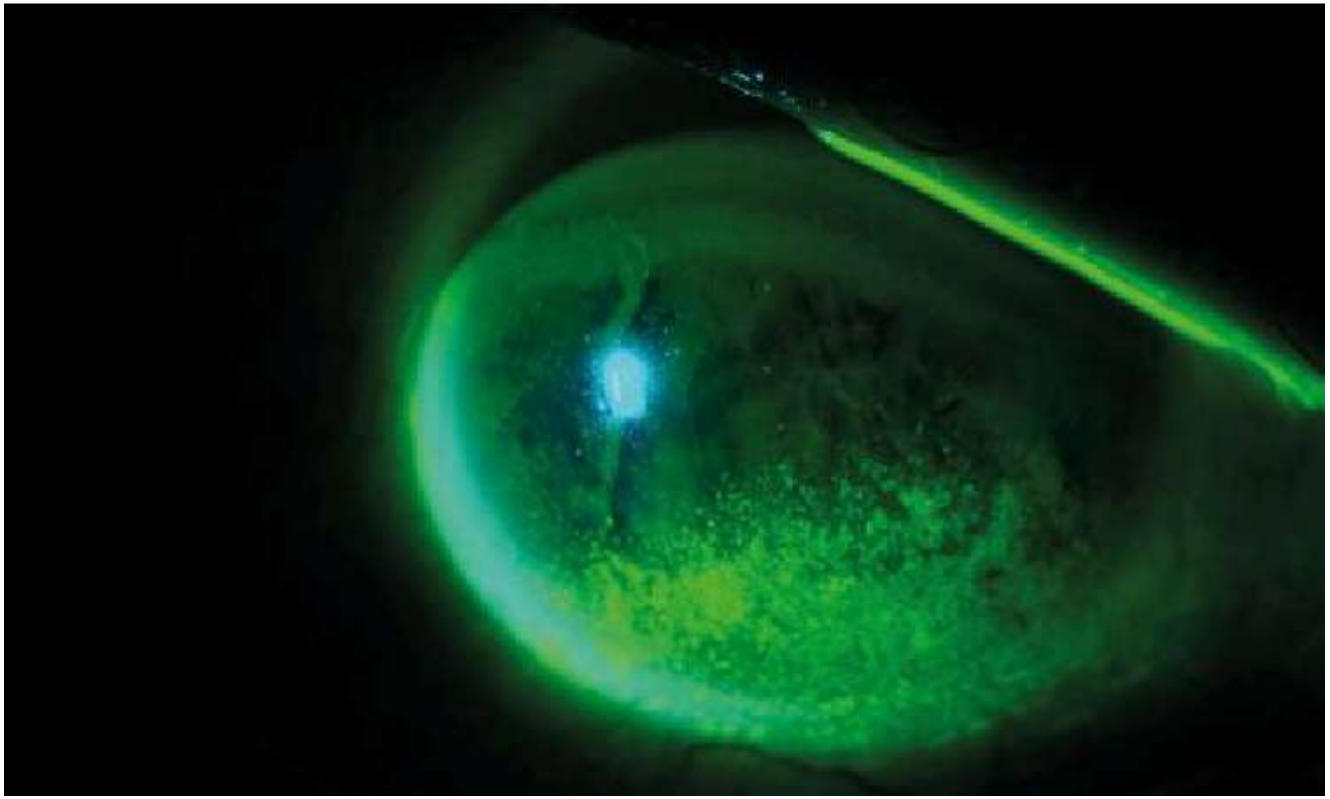


http://www.eyeworld.org/

Ectropion

Objective (continued):

- Exposure keratopathy
  - Inflammation of cornea caused by ***exposure***



https://www.semanticscholar.org/paper/How-to-care-for-a-patient%27s-eyes-in-critical-care-Marsden-Davies/5d60cf597faa244f0b81b778da66b6370d312b8a/figure/0

Ectropion

Objective (continued):

- Conjunctival injection
- Dilated blood vessels, which give red appearance to conjunctiva/sclera



http://www.djo.harvard.edu/files/2870_345.jpg

Ectropion

Objective (continued):

- Keratinization (2’ to conj drying)





https://webeye.ophth.uiowa.edu/eyeforum/atlas/pages/xerosis-conjuntivae.html

http://eyedepkgmu.blogspot.com/2014/11/kpro-keratinized-cornea-severe-dry-eyes.html

Ectropion



- Multiple causes of ectropion:
  - Acquired
    - Involutional (lid laxity → age)
    - Paralytic (nerve issues)
    - Cicatricial (due to scarring)
    - Mechanical (something is pushing or causing it)
    - Allergy (swelling)
  - Congenital (born that way)

Ectropion



Causes (Acquired)

- Involutional
  - Most common form of ectropion
  - Horizontal lid laxity
  - Usually due to age-related weakness
  - Begins medially → moves temporally

Ectropion

http://img.tfd.com/mosby/thumbs/50018X-fx5.jpg



Causes (Acquired)

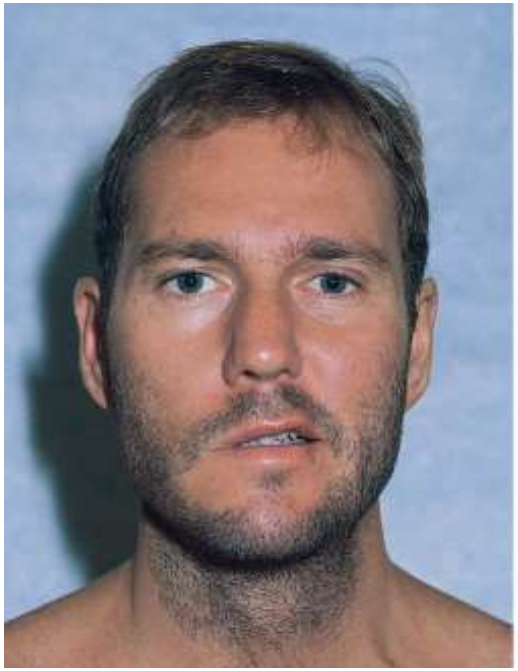
- Paralytic
  - CN VII palsy (ex. Bell's Palsy)
- The following should be documented:
  - Corneal integrity
  - Corneal sensation
  - Presence or absence of Bell's phenomenon
  - Degree of lagophthalmos

http://emedicine.medscape.com/article/1212398-clinical#showall

Ectropion

Ask patients with suspected facial nerve palsy:

- Wrinkle their forehead
- Puff out their cheeks
- Show their teeth rather than smile





Ectropion

- In patients with suspected facial nerve palsy:
  - Inferior scleral show should be distinguished from ectropion



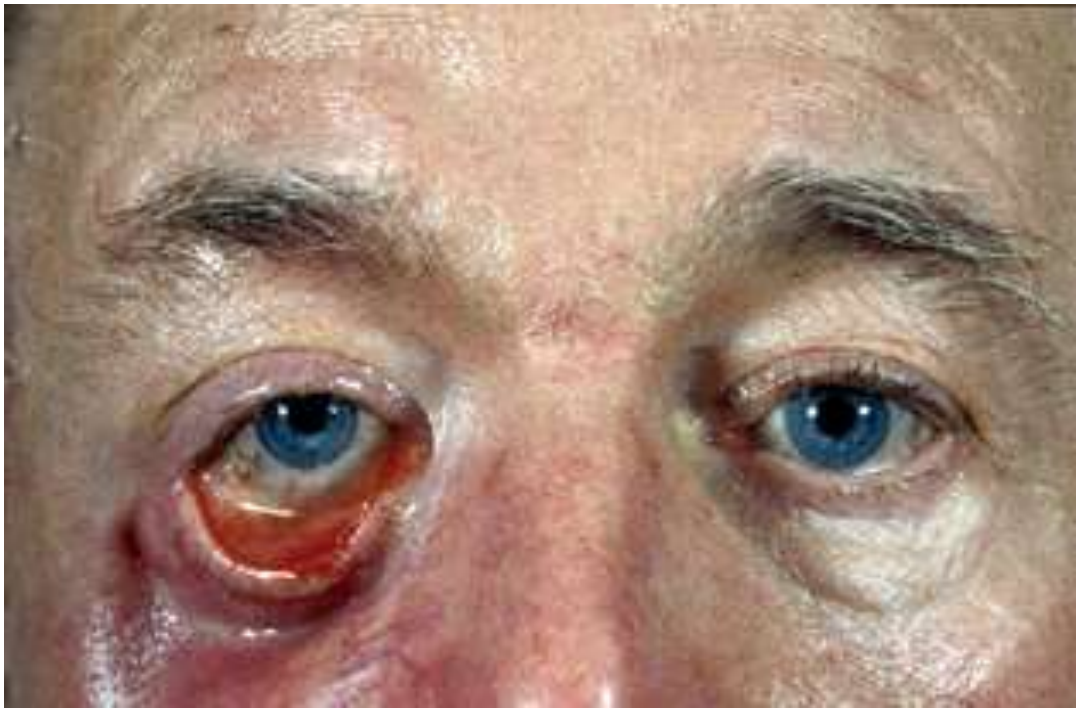
www.sarawakeyecare.com



http://www.graphicshunt.com

Ectropion

- Causes (Acquired)
  - Cicatricial (scarring)
    - Due to chemical burn, surgery, trauma, chronic dermatitis, surgery, fractures, other causes of scars



http://www.nature.com/eye/journal/v19/n8/images/6701667f3.jpg

Ectropion

- Causes (Acquired)
  - Mechanical
    - Due to herniated orbital fat, eyelid tumor and others
  - Allergic
    - Contact dermatitis



www.medflux.com

Ectropion





- Causes
  - Congenital
    - Rare
    - Associated with:
      - Blepharophimosis syndrome
        - Abnormal narrowness of palpebral fissure in horizontal direction
      - Microphthalmos
        - Congenital or developmental anomaly where globe abnormally small

Ectropion



http://pjmissions.blogspot.com/2015/05/empty-nest-syndrome.html

- Causes
  - Congenital (continued)
    - Associated with:
      - Buphthalmos
        - Enlarged globe resulting from increased IOP
        - Seen in patients with congenital glaucoma
      - Orbital cysts
        - Cyst around orbit, can cause pseudo-ptosis or other issues
      - Down syndrome
    - Occasionally congenital ectropion has paralytic cause

Ectropion

- Differentials:
  - Bell's Palsy
  - Floppy Eyelid Syndrome
  - Basal Cell Carcinoma
  - Squamous Cell Carcinoma
  - Ichthyosis
    - Uncommon group of skin disorders characterized by excessive amounts of dry surface scales

Ectropion

- Assessment
  - Take careful history
  - Slit-lamp exam – check for SPK, exposure keratopathy, etc.
  - Treatment based on cause and severity

Ectropion







Plan

- **<u>Medical Management</u>**
  - Lubricants
    - Drops
    - Ointments
  - Taping lids
    - Especially in patients with new-onset seventh nerve palsy
  - Lid manipulation
    - Patient instructed to wipe up and in when “drying” tears



http://www.cehjournal.org/images/ts100116.jpg

Ectropion

Plan

- **<u>Surgical Management (Blepharoplasty)</u>**
  - The surgical treatment depends on etiology
  - Temporary fixes:
    - Tarsorrhaphy
    - Electrocautery





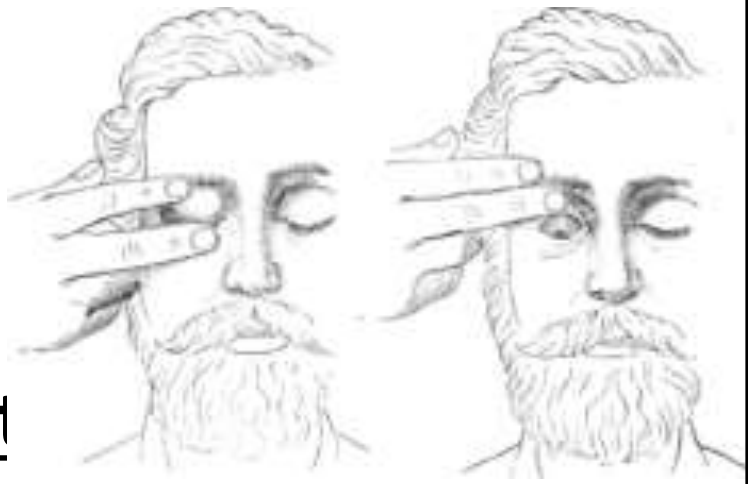
Ectropion



Plan

- **<u>Surgical Management (Blepharoplasty)</u>**
  - "Permanent" Fixes:
    - Involutional ectropion
      - Tarsal Strip
        - Few stitches;
        - Removed 7-14 days after surgery
        - Almost immediate resolution of condition

Ectropion



Plan

- **<u>Surgical Management (cont</u>**
  - "Permanent" Fixes:
    - Cicatricial ectropion following trauma or lid surgery
      - Digital massage may help stretch scar
      - Skin grafts are other option
        - Pre- or Post- auricular skin or eyelid skin*

http://chestofbooks.com/health/body/massage/Massage-Original-Swedish-Movements/XII-Massage-Of-The-Eye.html

Ectropion

TREATMENT	ECTROPION TYPE	SEVERITY OF ECTROPION
Lubrication	All	All (mainly mild-mod)
Tape	All	Moderate to severe
Tarsorrhaphy	All	Moderate to severe
Electrocautery	Involutional	Moderate to severe
Blepharoplasty	Involutional	Moderate to severe
Digital massage	Cicatricial	Mild to moderate
Skin graft	Cicatricial	Moderate to severe

Ectropion Summary



www.neweyelids.com/img

- OUTWARD turning of eyelid
- Symptoms:
  - Tearing, red eyes, sagging eyelid
- Different Causes (age, scarring, etc.)

- Different Treatments (patient education, lubricants, surgery)

Describe this Eye Lid



http://www.eyeatlas.com/box/28entropion.jpg

Entropion

- Definition: Inward turning of eyelid margin



Entropion



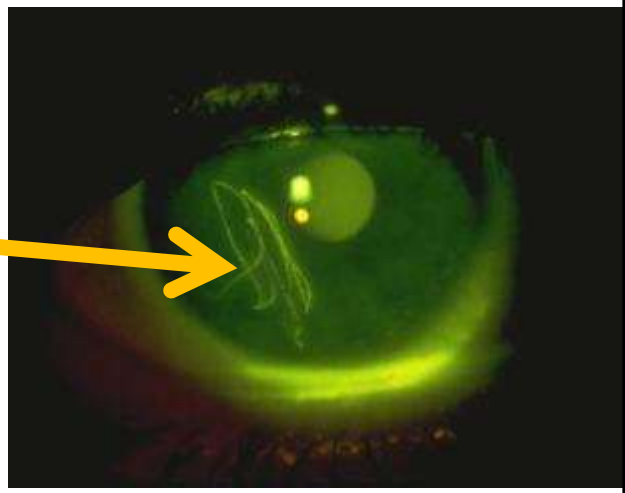
- Subjective:
  - Ocular irritation
  - Foreign-Body sensation (FB sensation)
  - Pain or no pain
  - Tearing
  - Red-eye
  - Light sensitivity (photophobia)
  - Possible decreased vision

http://www.perret-optic.ch/optometrie/symptomes_diagnostiques/symptomes/symptomes_image/opto_entropion.gif

Entropion

- Objective
  - The eyelid is turned in!
  - SPK/PEE (from exposure)
  - Foreign body tracking
  - Conjunctival injection
  - Corneal ulcer
  - Decreased corneal sensation
  - Corneal scarring if long-term





http://www.bausch.com/

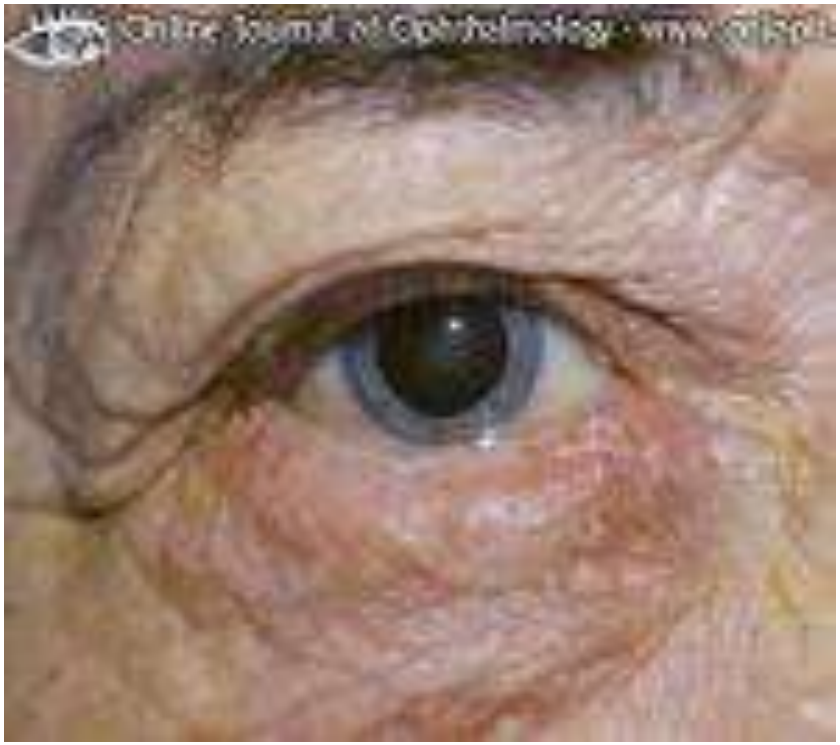


Entropion

- Causes
  - Involutional
  - Acute Spastic
  - Cicatricial
  - Congenital

Entropion

- Causes
  - Involutional
    - Laxity of lateral and medial canthal tendons
    - Over-acting/riding orbicularis oculi muscle





Entropion

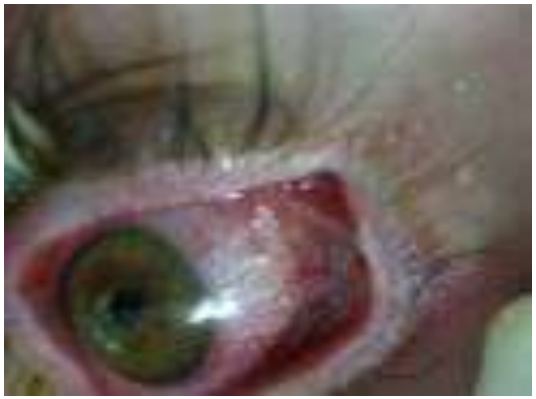
- Causes (continued)
  - Acute spastic entropion
    - Ocular irritation is predominate cause
    - Over-acting orbicularis oculi muscle
    - Most have concurrent involutional component.



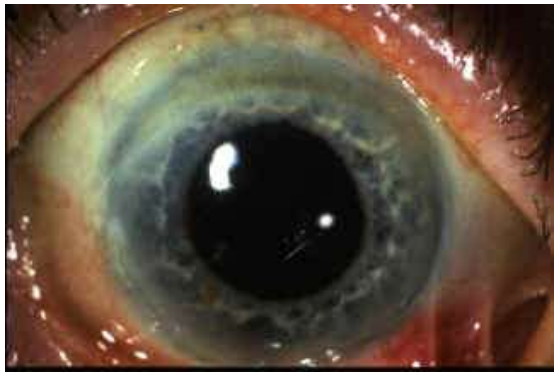
http://emedicine.medscape.com/article/1212456-clinical#showall

Entropion





- Causes (continued)
  - Cicatricial entropion
    - Result of scarring of palpebral conjunctiva
    - Digital eversion of eyelid margin is difficult
    - Scar tissue of conjunctiva usually result of:
      - Trauma
      - Infections
      - Chemical burns
    - Stevens-Johnson syndrome
    - Ocular cicatricial pemphigoid (OCP)



http://emedicine.medscape.com/article/1212456-clinical#showall

Entropion



- Causes (continued)
  - Congenital entropion
    - Very rare
    - Usually lower eyelid
    - Dysgenesis of lower eyelid retractors
      - Paucity of tissue vertically in posterior lamella of eyelid
      - Tarsal kink syndrome
        - Distortion of tarsal plate that may result in the in- or out-turning of lid

http://emedicine.medscape.com/article/1212456-clinical#showall

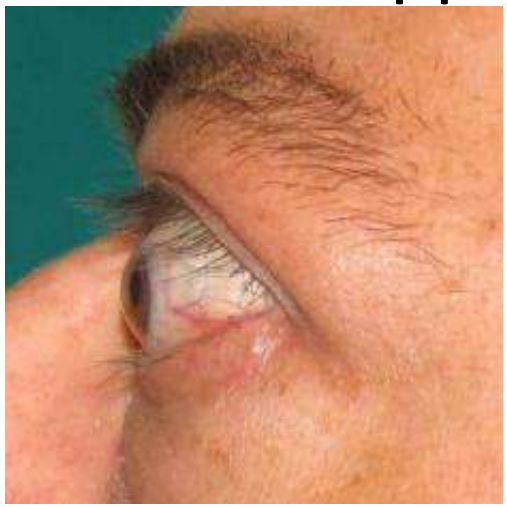
Entropion



- Other considerations
  - Epiblepharon
    - Pretarsal portion of orbicularis muscle and skin covering eyelid push eyelashes vertically or inwards
    - Eyelid margin is in normal position
    - Usually resolves spontaneously as face matures

Entropion

- Other considerations
  - Eyelid Retraction
    - May simulate entropion in both upper and lower lids
    - Lids exhibit normal apposition to globe



http://www.thyroidmanager.org/Chapter12/figures/figure4.jpg

Entropion

- Differential Diagnoses (short list)
  - Trichiasis – will soon discuss
  - Distichiasis
    - Lashes growing out of meibomian glands
    - Second row of lashes



http://www.nature.com/eye/journal/v18/n12/images/6701387f2.jpg

Entropion

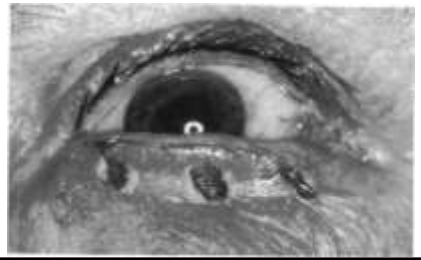
- Plan
  - **Medical Management**
    - Ocular lubrication and tear preparations
      - Protects ocular surface
    - Eyelid hygiene, antibiotics, and steroids
    - Bandage Contact Lens
    - Botulinum toxin (BOTOX®)

Entropion



- Plan
  - **<u>Surgical Management</u>**
    - Most common procedures:
      - Involutional entropion by repairing horizontal laxity
      - Cicatricial entropion
        - Mild cases - removal of scar
        - More extensive scarring may require grafts
    - Cautery is another option (temporary) – Poor aesthetics







Entropion Summary

- INWARD turning of eyelid
- Symptoms:
  - Tearing, red eyes, scarring
- Different Causes (age, spasm, scarring, etc.)



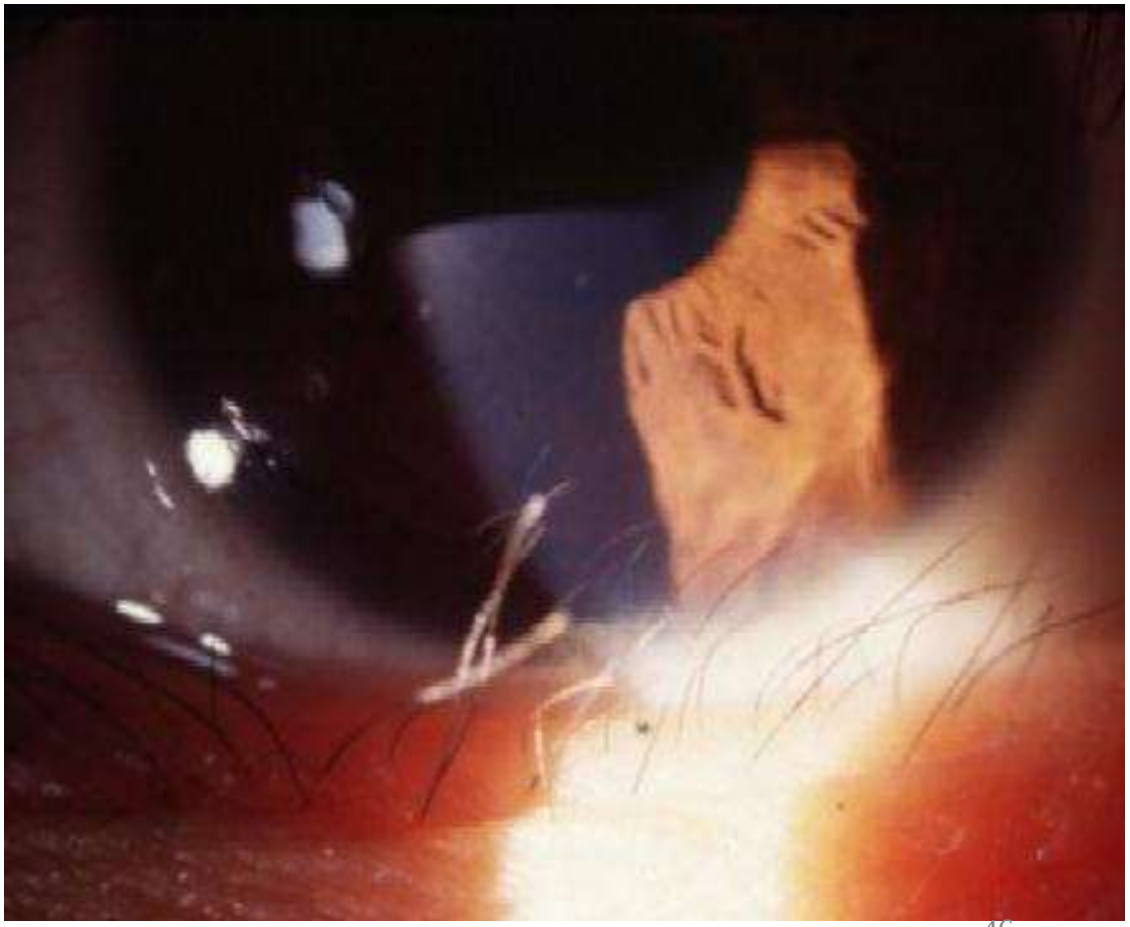
- Different Treatments (Lubricants to surgery)

Describe this Eye Lid



http://www.visualdxhealth.com/images/dx/webAdult/blepharitis_52998_lg.jpg

Trichiasis: Acquired misdirection of eyelashes toward the globe

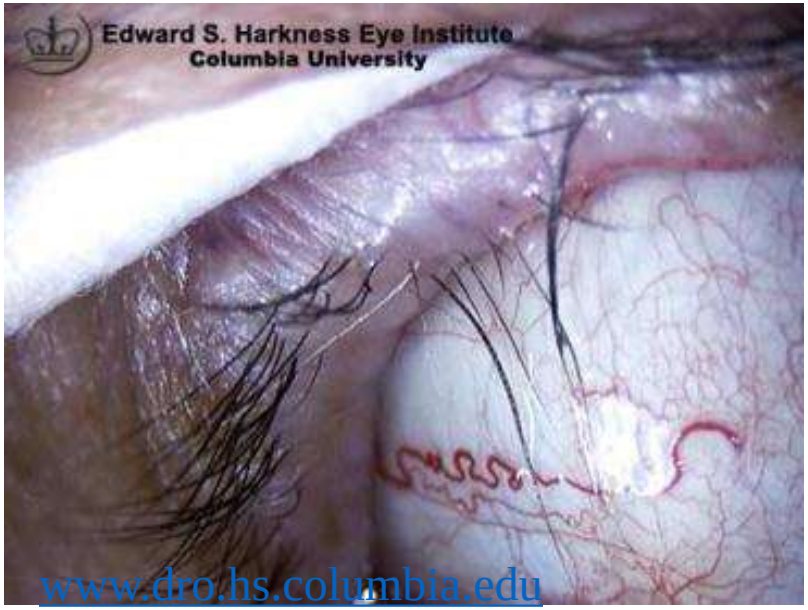


Trichiasis

- Subjective
  - FB sensation
  - Red eyes
  - Pain
  - Photophobia
  - Tearing

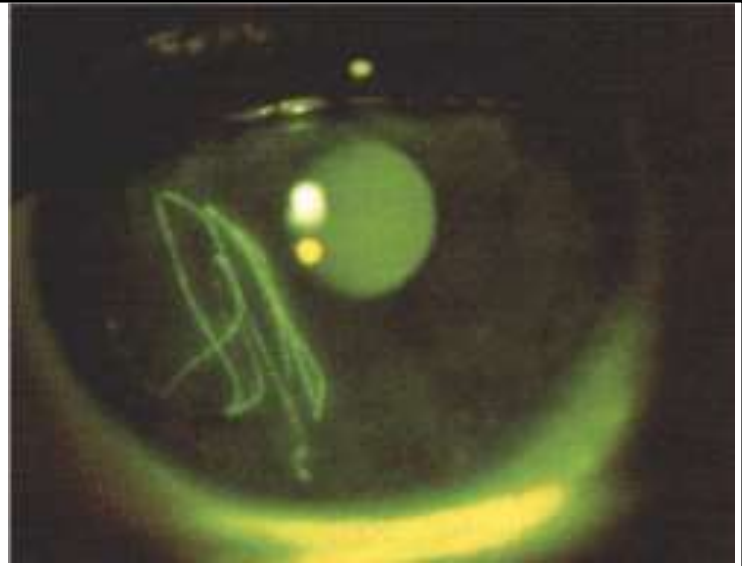
Trichiasis

- Objective
  - General slit lamp exam will reveal cause
  - Treatment based on cause of misdirection
  - Examine both upper and lower lids



www.dro.hs.columbia.edu

Trichiasis



www.webeye.ophth.uiowa.edu

- Objective
  - Look for
    - Conjunctival scarring
    - FB tracking on cornea (linear defects)
    - Entropion
    - Distichiasis
      - Lashes growing from meibomian gland orifices
    - Symblepharon formation

https://webeye.ophth.uiowa.edu/eyeforum/atlas/pages/symblepharon.html



Trichiasis



- Causes (not all inclusive)
  - Infectious
    - Blepharitis, HZ, Trachoma (chlamydia)
  - Autoimmune
    - Ocular cicatricial pemphigoid
  - Inflammatory
    - Stevens-Johnson syndrome
    - Vernal keratoconjunctivitis



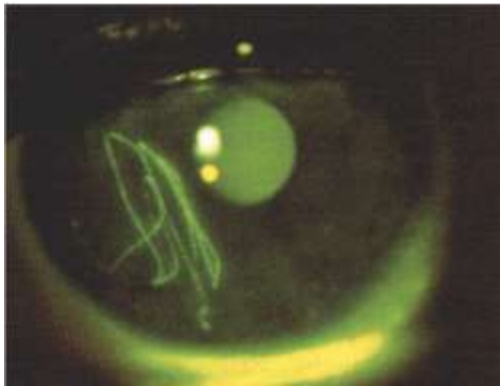
Trichiasis

- Causes (continued)
  - Trauma
    - Postsurgical
      - Floor fracture repair or blepharoplasty
      - After enucleation
      - After ectropion repair
    - Chemical
      - Alkali burns to eye
      - Medical drops (e.g. glaucoma drops)
    - Thermal burns to face/lids

Trichiasis



- Differentials
  - Blepharitis
  - Blepharospasm
  - Chemical burns
  - Cicatricial Pemphigoid (OCP)
  - Corneal Abrasion
  - Corneal Foreign Body
  - Distichiasis
  - Entropion
  - Red eye
  - Stevens-Johnson Syndrome (SJS)
  - Trachoma



Trichiasis

- Plan: **Medical Management**
  - Ocular Lubricants
  - Treat underlying disease process

Trichiasis

- Plan: **<u>Surgical Management</u>**
  - Technique dependent on cause
  - Two categories of procedures:
    1. Lash/follicle destroying
    2. Lash/follicle repositioning

Trichiasis

http://www.nature.com/eye/journal/v20/n8/thumbs/6702066f1th.jpg



- Plan: **Surgical Care**
  - Lash and follicle **destruction**
  - Preferred for segmental or focal trichiasis.
    - Epilation – usually temporary
      - When lash grows back - often short and irritating
    - Electrolysis
      - Can be effective
      - Often painful for patient and tedious for surgeon
    - Radiofrequency ablation
      - Extremely effective
    - Cryosurgery - effective, but many potential complications
    - Wedge resection of lid
      - Requires full-thickness resection of lid margin



Trichiasis

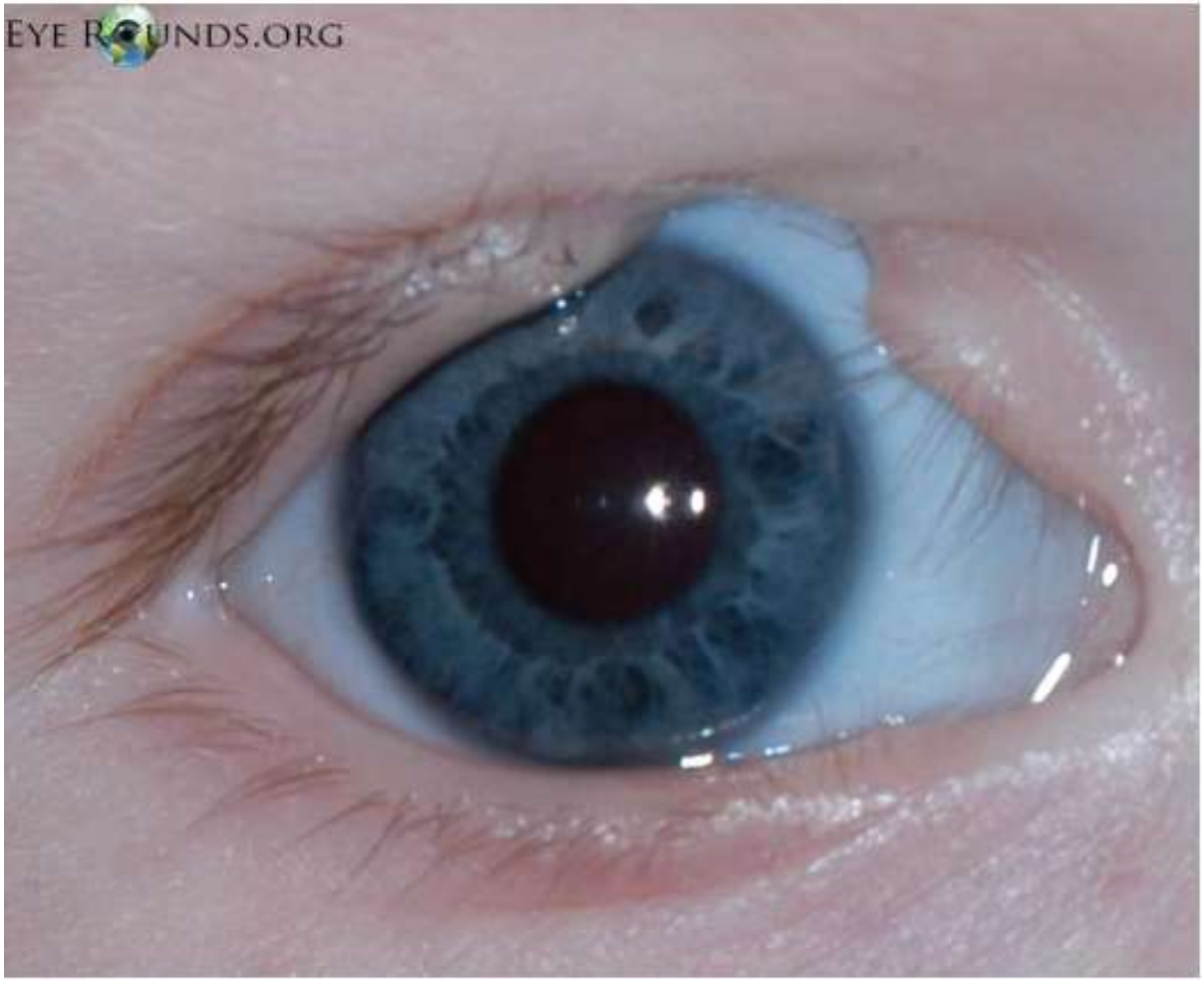
- Plan: **<u>Surgical Care</u>**
  - Lash and follicle **repositioning**
  - Directed toward anatomical cause of problem
    - Elevated risk of entropion

Trichiasis Summary

- Misdirection of lashes towards globe
- Causes irritation
- Determine cause
- Preferred treatment option: Epilation



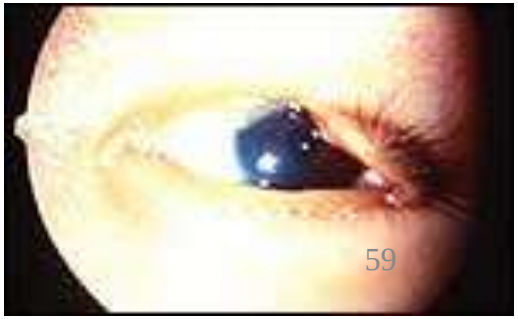
Describe this Eye Lid?



https://webeye.ophth.uiowa.edu/eyeforum/atlas/pages/eyelid-coloboma.htm

Eyelid Coloboma: full-thickness defect of the eyelid

- Triangular defect
  - Base of notch primarily at lid margin
- It primarily affects upper lid
  - Between inner to middle third of lid
- If in the lower lid
  - Between middle to outer third of lid



Coloboma

- Rare (~1/10,000)
- Primarily congenital, but may be 2’ to trauma
- Various structures affected
- Can be associated with cryptophthalmos
  - Failure of eyelid formation

Coloboma

- Want to take good history
  - Perinatal and pregnancy history
  - Family history of congenital eyelid colobomas or other congenital anomalies
  - History of other current birth defects
  - Pediatric review of systems
  - History of progressive corneal problems

Associated Ocular Defects



Structure	Ocular condition(s)
Eyelids	Trichiasis Dermoid tumors
Conjunctiva	Injection Symblepharon Malformation of the caruncle
Cornea	Exposure keratopathy Corneal scarring
Iris	Coloboma
Lens	Cataract (anterior) Subluxation
Sclera	Epibulbar Dermoid Tumor

Coloboma

- Plan: **Medical Management**
  - Bandage contact lenses
  - Lubricants
  - Bedtime taping/patching



Essentialy: Corneal Protection

Coloboma

- Plan: **Surgical Management**
  - Large lid coloboma - immediate surgical closure
  - 2-stage reconstruction may be required
    - Tarsal reconstruction with skin graft

Coloboma Summary

- Rare
- Primarily congenital
- Full-thickness eyelid defect
- Protect cornea and assist aesthetically
- Surgery usually indicated

Describe these Eye Lids



Epicanthus

- Definition: Crescent fold of skin that extends from side of nose to lower lid and partially covers inner canthus
- Bilateral
  - Although may be asymmetrical
- Normally, asymptomatic



Epicanthus



- Objective
  - Pseudo-esotropia
    - False inward turning of eyes
  - Infants & toddlers
    - Most prominent:
      - 3-6 months of age
      - Asian descent
  - Common in those with Trisomy 21
  - Clinical epicanthus = failure of epicanthus to regress

Epicanthus

- Systemic Disorders associated w/Epicanthus
  - Chromosomal Disorders

    Turner's syndrome
    Klinefelter's syndrome
    Deletion of short term arm of chromosome 5
    Deletion of long arm of chromosome 13
    Deletion of short arm of chromosome 18
    Trisomy 10q+
    Trisomy 14q+
    Trisomy 18
    Trisomy 21
    Trisomy 22
  - Nonchromosomal systemic disorders

    Ehlers-Danlos syndrome
    Smith-Lemli-Opitz syndrome
    Cerebrohepatorenal (Zellweger) syndrome

Epicanthus

- Plan
  - No Treatment
  - Patient/parent education and reassurance
  - Infants re-evaluated every 6 months
    - True esotropia (ET) (especially accommodative ET) may develop
  - If visual field affected, surgery can be performed

Epicanthus Summary

- Crescent shape folds that cover inner canthus and give appearance of esotropia
- Bilateral
- Common in infants/toddlers
- No Treatment



Describe these Eye Lids



http://www.saunalahti.fi/kymensil/kuvat/roluo.jpg

Dermatochalasis

- Definition: Redundant, lax eyelid skin overhanging the margin



Dermatochalasis

- Mainly seen in elderly
- Very common condition
- Onset in 40’s and progresses



Dermatochalasis



- Causes:
  - Gravity
  - Loss of elastic tissue in skin, and
  - Weakening of connective tissues

- More common in upper eyelids but can be seen in lower eyelids

Dermatochalasis



- Subjective
  - Functional and/or cosmetic problem
    - In addition, patients may note:
      - Foreign body sensation
      - Entropion of upper eyelid
      - Ectropion of lower eyelid
      - Blepharitis
      - Dermatitis

Dermatochalasis



- Subjective (Cosmetic):
  - Patients note:
    - Fullness or heaviness of upper eyelids
    - “Bags” in lower eyelids
    - Wrinkles in lower eyelids and lateral canthus

Epicanthus

- Objective
  - Eyelid skin should be evaluated carefully
    - Redundancy
    - Thickness
    - Inflammation
    - Any lesions
  - Levator muscle intact



Dermatochalasis

- Objective (continued)
  - Eyelid margin position also should be noted
  - Lid should be 1-2 mm below superior limbus
  - Upper eyelid crease should be measured
    - Normally 8-12 mm above lid margin
    - Higher in women

Dermatochalasis

- Causes
  - Aging*
  - Trauma
  - Periorbital edema
    - Associated with chronic renal insufficiency
  - Chronic dermatitis
  - TED
    - Associated with infiltration of orbital fat and EOM's

Dermatochalasis

- Causes (continued)
  - Floppy eyelid syndrome
    - Associated with severely redundant skin and lax lids
      - Skin and muscles affected
      - Tarsal plate develops rubbery consistency
    - Associated with Obstructive Sleep Apnea (OSA)

Dermatochalasis



- Differential Diagnoses:
  - Entropion
  - Floppy Eyelid Syndrome
  - **Blepharochalasis** - distinct from dermatochalasis
    - Rare disorder that typically affects the upper eyelids
    - Characterized by intermittent eyelid edema
    - Results in relaxation of eyelid tissue and atrophy
    - Unilateral in ~50%





Dermatochalasis

- Differential Diagnoses (continued):
  - Steatoblepharon
    - Herniation of orbital fat in upper or lower eyelids
  - Ptosis





Dermatochalasis

- Plan
  - Assess functional impact with Visual Field
  - Pts with blepharitis require lid hygiene and antibiotics
  - Pts with dermatitis may benefit from steroid ointment
  - Treatment is SURGICAL





Dermatochalasis

- Plan
  - **<u>Surgical Management</u>**:
    - Upper eyelid blepharoplasty after:
      - Careful Hx
      - Evaluation (VA, VF, etc.)





THANK YOU!